

Lab 4 Solutions Group(1-8)

Secret Key :

```
Letters = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
```

```
s = input().upper()
#s--> input string
k = int(input())
out = ""
for ch in s:
    if ch in Letters:
        idx=Letters.find(ch)
        new_idx = (idx+k)%26
        out += Letters[new_idx]
    else:
        out +=ch

print(out)
```

John's Message 1 :

```
##Approach two --> with slicing
```

```
input_string = input()
print(input_string[2::3])
```

```
##Approach one --> without slicing
```

```
toPrint = ""
```

```
input_string = input()
```

```
pos_i = 2
```

```
while(pos_i < len(input_string)):
    toPrint += input_string[pos_i]
    pos_i +=3
```

```
print(toPrint)
```

Moving On :

```
inp = input().split(" ")
x = inp[0]
y = inp[1]
alphabets = "abcdefghijklmnopqrstuvwxyz"
ans = 0
for c in alphabets :
    f1 = x.count(c)
    f2 = y.count(c)
    ans += abs(f1-f2)
    # print(abs(f1-f2))
print(ans)
```

John's Message II

using slicing -->

```
s = input().split()
```

```
print(s[1][int(s[0])-1:int(s[0])])
```

Incomprehensible, what ?

```
def cum_sum(lis):
    out = ""
    c_sum = 0
    for num in lis:
        c_sum += num
        out += str(c_sum)

    return out[:-1]
```

```
lis = [int(x) for x in input().split()]

print(cum_sum(lis))
```

Count Vowels

```
s = input()

arr = [0, 0, 0, 0, 0]

for i in s:
    if i == 'a':
        arr[0] += 1
    elif i == 'e':
        arr[1] += 1
    elif i == 'i':
        arr[2] += 1
    elif i == 'o':
        arr[3] += 1
    elif i == 'u':
        arr[4] += 1
```

```
print('a:', arr[0])
print('e:', arr[1])
print('i:', arr[2])
print('o:', arr[3])
print('u:', arr[4])
```

Help Sam Again !!

```
string = input()
n = len(string)
mx = 0
for i in range(n):
    f = string[i]
```

```
s = string[i]
j = i
while j<n :
    if(string[j]==f):
        j+=1;
        continue
    if(f==s):
        s = string[j];
        j+=1;
        continue;
    if(string[j]==s):
        j+=1;
        continue;
    break;
    j+=1;
mx = max(mx,j-i)
print(mx)
```